# Mechanical Design & Simulation - Task 1 Report

## Objective

Design a 3D connecting rod model using SolidWorks and assign a suitable engineering material.

## Software

SolidWorks

| Feature | Dimension (mm) |
|---|---|
| Big End OD | 60 |
| Big End ID | 30 |
| Small End OD | 35 |
| Small End ID | 18 |
| Center Distance | 120 |
| Rod Width | 20 |
| Thickness | 15 |
| Fillet Radius | 5 |

## Material

Mild Steel: High strength, durable, commonly used for connecting rods, easy to machine.

## Design Procedure

1. Create a new Part.
2. Sketch concentric circles for big and small ends.
3. Connect with tangent lines.
4. Fully define sketch.
5. Extrude 15 mm.
6. Apply 5 mm fillets.
7. Assign Mild Steel.
8. Save and capture Front, Top, Isometric and Rendered views.

## Observations

The model was created successfully. Fillets improve stress distribution and appearance. Mild steel provides suitable mechanical properties.

## Conclusion

The connecting rod CAD model satisfies the required dimensions and is ready for rendering and further analysis.